

Answers:

1. True/False: Games typically use square matrices for most calculations.
 - a. True
2. True/False: Matrices always have values ranging from 0 to 1.
 - a. False, they're not limited to any particular range.
3. True/False: A 1x2 matrix consists of 1 row and 2 columns.
 - a. True
4. True/False: Vectors can be considered special cases of matrices.
 - a. True, we haven't covered this yet, but they absolutely can be considered like a matrix with either a single row or single column.
5. True/False: A 3x3 matrix has 3 rows and 3 columns.
 - a. True
6. True/False: If you're given an MxN matrix, the M represents the # of columns, while N represents the # of rows.
 - a. False, the first number is rows, the 2nd is columns.
7. True/False: Most matrix math in games revolves around linear transformations.
 - a. True
8. True/False: A square matrix is one that has the same number of rows and columns.
 - a. True
9. True/False: A 3x4 matrix has 4 rows and 3 columns.
 - a. False, it has 3 rows and 4 columns. Generally, you read it as "rows x columns".
10. Give an example of a 3x2 matrix.
 - a. Really, anything here works, as long as you demonstrate it has 3 rows and 2 columns

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

11. What size is this matrix?
 - a. 3x3
12. What size is this matrix?
 - a. 2x3
13. What size is this matrix?
 - a. 4x2